INCH-POUND
MIL-V-24722/1(SH)
15 JUNE 1990

MILITARY SPECIFICATION SHEET

VALVE AND SUBPLATE, SERVO FLOW-CONTROL, ELETROHYDRAULIC FOR 0 TO 50 GALLONS PER MINUTE (GPM) AND 0 TO 80 GPM FLOW RANGES

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-V-24722.

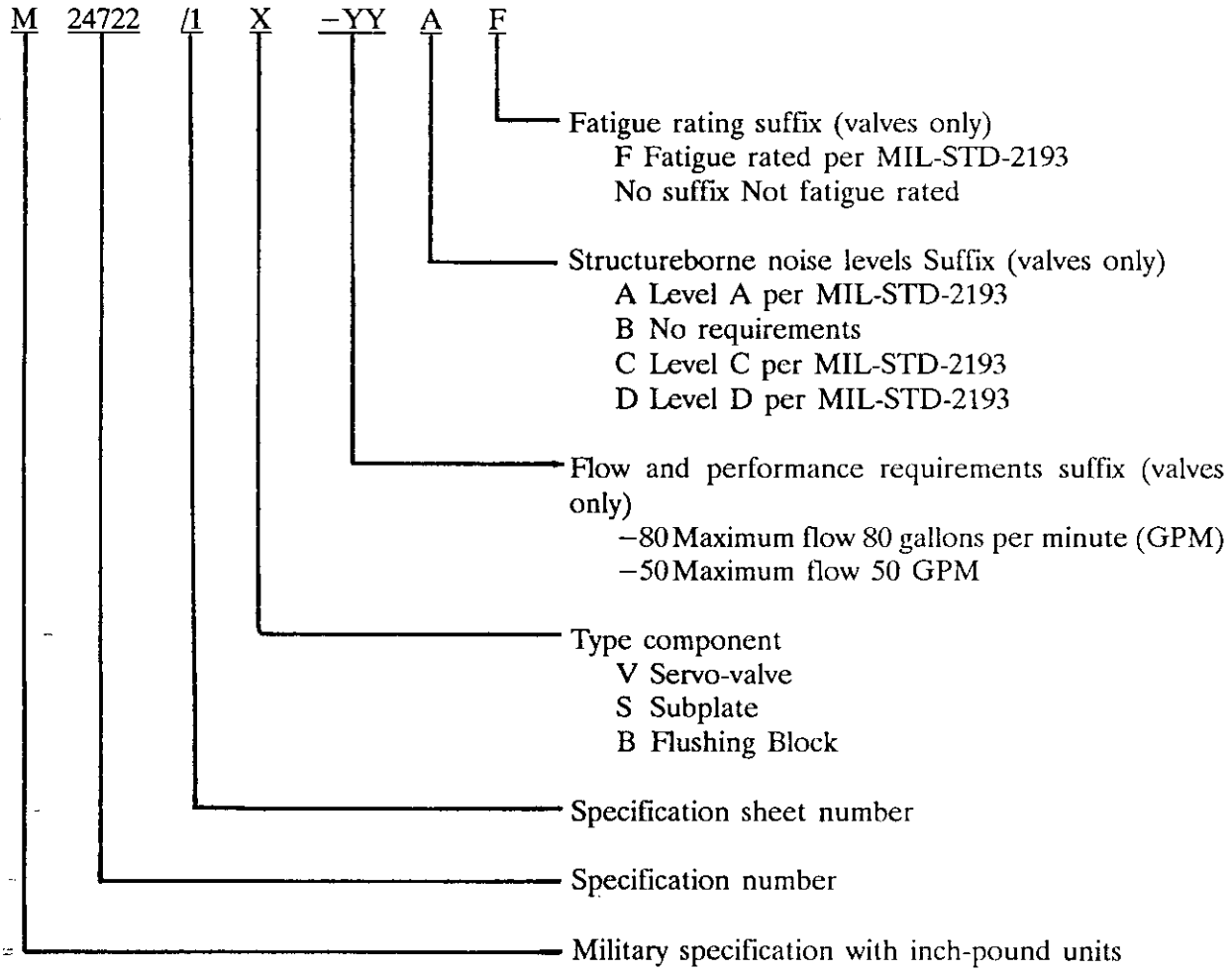
1. GENERAL

1.1 Part or identifying number. Servo control valves and subplates covered by this specification sheet shall be designated using Part or Identifying Numbers (PIN) as follows:

AMSC N/A FSC 4810

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Examples:	M24722/1V-8OAF	A servo-valve with maximum flow of 80 gpm, level A noise requirements per MIL-STD-2193, and fatigue rated
	M24722/1V-80B	A servo-valve with maximum flow of 50 gpm, no noise or fatigue requirements
	M24722/1S	A subplate.

1.2 **Mechanical configuration.** The subplate configuration of this valve shall be as shown in figures 1 and 2. Envelope dimensions of the valve/subplate assembly shall not exceed the envelope requirements of figure 3. The manifold pattern of this valve shall be as shown in figure 4.

1.3 **Design operating pressure.** The design operating pressure of this valve shall be 3000 pounds per square inch (psi) gauge.

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1.4 Electrical connector. The MIL-C-5015 five-pin receptacle connector shall be in accordance with MS3102R18-11P. Mechanical crimping in lieu of soldered connections is acceptable.

1.5 Torque motor current. Rated current applied to the torque motor from the amplifier shall be plus or minus 24 milliamperes (mA) for minus 10 flow rated valves and plus or minus 100 mA for minus 15 flow rated valves about zero quiescent current. A greater current flowing from pin C to pin A than from pin C to pin D shall cause flow out of port C1. A greater current flowing from pin C to pin D than from pin C to pin A shall cause flow out of port C2.

1.6 Torque motor resistance. The measured pin-to-pin resistance shall be 730 plus or minus 10 ohms across each coil for minus 10 flow rated valves and 220 plus or minus 11 ohms across each coil for minus 15 flow rated valves. The resistances of each pair of coils shall be matched to within 1 percent or less.

1.7 Pilot stage return porting. Pilot valve and main stage return flows shall be ported separately to the subplate.

1.8 Pilot stage output test ports. The valve shall have normally plugged PV1 and PV2 test ports that can be opened to permit connection of gages to measure control pressure at the driving pistons of the main stage. These ports shall use a gasket seal straight thread boss in accordance with MS16142.

1.9 Fluid connections. Pressurized fluid connections between internal portions of components and between valve and subplate shall be designed using straight-through quills having static o-ring seals.

2. TEST CONDITIONS

2.1 Test fluid. Hydraulic fluid for first article testing and quality conformance testing of valves and subplates conforming to this specification sheet shall be in accordance with MIL-L-17331, 2190 TEP.

2.2 Fluid contamination level. Valves and subplates shall perform in accordance with the first article testing and quality conformance test requirements specified herein when operated in a hydraulic system which shall not be cleaner than NAS 1638 class 8.

2.3 Set conditions. The valve shall be quality conformance tested within the table II specified limits for pressure, temperature, flow, and torque motor signal command. Valve and subplate first article testing shall be performed in accordance with the limits and sequences of both tables I and II.

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3. FIRST ARTICLE TESTING

All of the examination and test condition requirements of tables I and II apply to valves and valve-subplate assemblies. A restricted number of tests, as noted in tables I and II, shall be conducted on subplates when tested by themselves.

3.1 Vibration. The valve shall be filled with fluid and subjected to a variable frequency test for type 1 equipment for vibrations up to 40 hertz (Hz) in accordance with MIL-STD-167-1. Between 33 and 40 Hz, table amplitudes of 0.007 plus or minus 0.002 inches (single amplitude) shall be used.

3.2 Contamination sensitivity. The contamination sensitivity test shall be conducted in accordance with MIL-STD-2193 with contaminant which conforms to the requirements of AC Fine Test Dust.

4. QUALITY CONFORMANCE TESTING

All of the examination and test condition requirements of table II apply to valves and valve-subplate assemblies. A restricted number of tests, as noted in table II, shall be conducted on subplates when tested by themselves.

4.1 Null lap. No-flow cylinder port pressures shall be within 1500 plus or minus 1000 psi at valve null under the conditions specified in table II.

4.2 Hysteresis. The valve hysteresis shall not exceed plus or minus 3 percent of the maximum rated torque motor current.

4.3 Threshold. Flow threshold shall not exceed plus or minus 1.5 percent of the maximum rated torque motor current.

4.4 Pressure gain. The pressure gain shall meet or exceed 20 percent of maximum load pressure drop, as defined in SAE ARP 490, for each 1 percent change of signal input within the plus or minus 40 percent load pressure drop range.

4.5 Amplitude ratio. The valve amplitude ratio values shall be flat within plus or minus 3 decibels (dB) out to 1 Hz for minus 10 flow rated valves and 4 Hz for minus 15 flow rated valves when a sinusoidal torque motor input command of not less than plus or minus 25 percent of rated current is applied.

4.6 Phase lag. The valve phase lag shall be less than 90 degrees at 1 Hz for minus 10 flow rated valves and less than 90 degrees at 4 Hz for minus 15 rated valves.

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4.7 Noise. The structureborne noise measurements shall be made with the subplate attached and mounted by any method (solid, shock cord, or isolation mounts). The structureborne noise-measuring accelerometer shall be mounted on the main return port flange. Valve ports C1 and C2 shall be connected with a hose. The structureborne noise test shall be run at the test points and valve operating conditions specified in table II. Test point 1, as defined in table I, shall be run by increasing flow linearly from 0 to 20 gpm over a 2 minute period then returning to zero flow in the same fashion. Ambient noise measurements on the subplate, including instrumentation error, shall not exceed 55dB or 10dB below the required acceptance levels, whichever is larger. The valve, including the subplate, shall not exceed the level A structureborne noise requirement in accordance with MIL-STD-2193 for solidly mounted equipment.

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TABLE I. *Required examination and test conditions – first article inspection.*

Test description	MIL-V-24722 requirements		M-24722/1 design and test require-ments	Test conditions[1]						Fluid flow[2] Set (S) Response (R)	Applicable to subplate
				Fluid pressure (psi)				Fluid temp (°C)	Torque M current (percent)		
	Design	Test		Supply (P)	Return (R)	Pilot S (PP)	Pilot R (PR)				
GROUP 1											
Perform general examinations and tests of tables I and II of the general specification											
GROUP 2											
Perform examinations and tests per table II of this specification sheet											
GROUP 3											
Flow gain-low fluid temp	3.5.4.2	4.3.8		(P–R=1200±30)		3000±50	Rsvr[3]	32±1			No
–50 flow rated			Fig 6						Fig 5	(R)	
–80 flow rated			Fig 5						Fig 6	(R)	
Flow gain-hi fluid temp.	3.5.4.2	4.3.8		(P–R=1200±30)		3000±50	Rsvr	53 min			No
–50 flow rated			Fig 6						Fig 5	(R)	
–80 flow rated			Fig 5						Fig 6	(R)	
Magnetic Field Effects	3.5.2.9	4.3.7		(P–R=1200±30)		2900±10	Rsvr	43±5			No
–50 flow rated			Fig 6						Fig 5	(R)	
–80 flow rated			Fig 5						Fig 6	(R)	
Endurance	3.5.6	4.3.20		2900±100	150±3	2900±100	150±50	43±5	±100	(S)	No
Internal leakage[5]	3.5.3.6	4.3.11		2900±100	Open	2900±100	Open[4]	53 min	0	2050cc[6]	No
Flow gain[5]	3.5.4.2	4.3.8		(P–R=1200±30)		2900±100	Rsvr	53 min	±100	(R)	No
Proof pressure[5]	3.5.1.1.1	4.3.2.2		4500 min	4500 min	4500 min	Rsvr	43±5	0	Zero	No

See footnotes on last page of table.

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TABLE I. *Required examination and test conditions – first article inspection – Continued.*

Test description	MIL-V-24722 requirements		M-24722/1 design and test require-ments	Test conditions[1]						Fluid flow[2] Set (S) Response (R)	Applicable to subplate
				Fluid pressure (psi)				Fluid temp (°C)	Torque M current (percent)		
	Design	Test		Supply (P)	Return (R)	Pilot S (PP)	Pilot R (PR)				
Shock	3.6.3	4.3.21									No
Vibration	3.6.4	4.3.22	3.1								No
Contamination sensitivity	3.5.3.7	4.3.24	3.2	(P–R=1200±30)		2900±100	Rsvr	43±5	±100	(S)	No
Noise	3.5.3.4	4.3.27	3.1	(P–R=1200±30)		2900±100	Rsvr	43±5		(S)	No
Salt Fog	3.6.5	4.3.25									Yes
Maintenance	3.9	4.3.19									No
GROUP 4											
Repeat examinations and tests per table II of this specification sheet											
GROUP 5											
Burst		4.3.28		7500±100	7500±100	7500±100	1500±100	43±5		Zero	Yes

[1]Tests which have blank rows in the "test conditions" columns are static, nonoperational tests which do not require a flow bench.
[2]Fluid flow is indicated to be either a set test condition (fixed or variable) or a dependent response with acceptance criteria specified in either the specification sheet (paragraph or figure) or the general specification.
[3]Rsvr – Valve R port connected to test bench hydraulic reservoir. Valve PR port joined to valve R port.
[4]Open – Valve R port exhausts to atmosphere for fluid sample collection. Valve PR port either exhausts to atmosphere or is separately returned to test bench hydraulic reservoir.
[5]These tests shall be performed at 25, 50, and 75 percent intervals per endurance test requirements of the general specification.
[6]Maximum allowable combined leakage from return port (R) and pilot return (PR) after $2x10^6$ cycles.

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TABLE II. *Required examination and test conditions – quality conformance (also first article test groups 2 and 4 as required).*

Test description	MIL-V-24722 requirements		M-24722/1 design and test require-ments	Test conditions[1]						Fluid flow[2] Set (S) Response (R)	Applicable to subplate
				Fluid pressure (psi)				Fluid temp (°C)	Torque M current (percent)		
	Design	Test		Supply (P)	Return (R)	Pilot S (PP)	Pilot R (PR)				
Proof pressure	3.5.1.1.1	4.3.2.2	1.3								
Test point #1				4500 min	4500 min	4500 min	Rsvr[3]	43±5	0	Zero	Yes
Test point #2				Resv	Resv	Resv	500	43±5	0	Zero	Yes
External leakage	3.5.1.1.3	4.3.2.2		4500 min	4500 min	4500 min	Rsvr	43±5	0		Yes
Polarity/run-in	3.5.4.1	4.3.3	1.5	2900±100	Resv	2900±100	Rsvr	43±5	±100	(S)	Yes
Flow gain	3.5.4.2	4.3.8		(P–R=1200±30)		3000±50	Rsvr	43±1			
–50 flow rated			Fig 6						Fig 5	(R)	No
–80 flow rated			Fig 5						Fig 6	(R)	No
Rated flow	3.5.3.1	4.3.9		(P–R=1200±30)		2900±100	Rsvr	43±2	±100		
–50 flow rated										50±3 gpm	No
–80 flow rated										80±5 gpm	No
Null lap	3.5.4.4	4.3.12	4.1	2900±100	Resv	2900±100	Open[4]	43±2	Pres null	C1 & C2 blocked	No
Hysteresis	3.5.4.5	4.3.13	4.2	2900±100	Resv	2900±100	Open	43±2	±100 var	Variable	No
Threshold	3.5.4.6	4.3.14	4.3	2900±100	Resv	2900±100	Open	43±2	Pres null ±1.5	Zero	No

See footnotes on last page of table.

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TABLE II. *Required examination and test conditions – quality conformance (also first article test groups 2 and 4 as required) – Continued.*

Test description	MIL-V-24722 requirements		M-24722/1 design and test requirements	Test conditions[1]						Fluid flow[2] Set (S) Response (R)	Applicable to subplate
				Fluid pressure (psi)				Fluid temp (°C)	Torque M current (percent)		
	Design	Test		Supply (P)	Return (R)	Pilot S (PP)	Pilot R (PR)				
Pressure gain	3.5.4.7	4.3.15	4.4	2900±100	Resv	2900±100	Open	43±2	Press null ±3.0	Zero	No
Internal leakage	3.5.3.6	4.3.11		2900±100	Open[2]	2900±100	Open	53 min	0	1640cc[5]	No
Amplitude ratio	3.5.5.1	4.3.16	4.5	2900±100	150±50	2900±100	150±50	43±2	±25 min	±25%	No
Phase lag	3.5.5.2	4.3.17	4.6	2900±100	150±50	2900±100	150±50	43±2	±25 min	±25%	No
Noise[6]	3.5.3.4	4.3.18	4.7								Yes
Test condition #1				(P–R=2800±100)		2900±100	Rsvr	54±2	As req'd	0 to 25%	
Test condition #2				(P–R=2500±100)		2900±100	Rsvr	54±2	As req'd	100%±6%	
Test condition #3				(P–R=1200±100)		2900±100	Rsvr	54±2	As req'd	100%±6%	

[1]Tests which have blank rows in the "test conditions" columns are static, nonoperational tests which do not require a flow bench.

[2]Fluid flow is indicated to be either a set test condition (fixed or variable) or a dependent response with acceptance criteria specified in either the specification sheet (paragraph or figure) or the general specification.

[3]Rsvr – Valve R port connected to test bench hydraulic reservoir. Valve PR port joined to valve R port.

[4]Open – Valve R port exhausts to atmosphere for fluid sample collection. Valve PR port either exhausts to atmosphere or is separately returned to test bench hydraulic reservoir.

[5]Maximum allowable combined leakage from return port (R) and pilot return (PR).

[6]Perform as required by quality conformance sampling plan.

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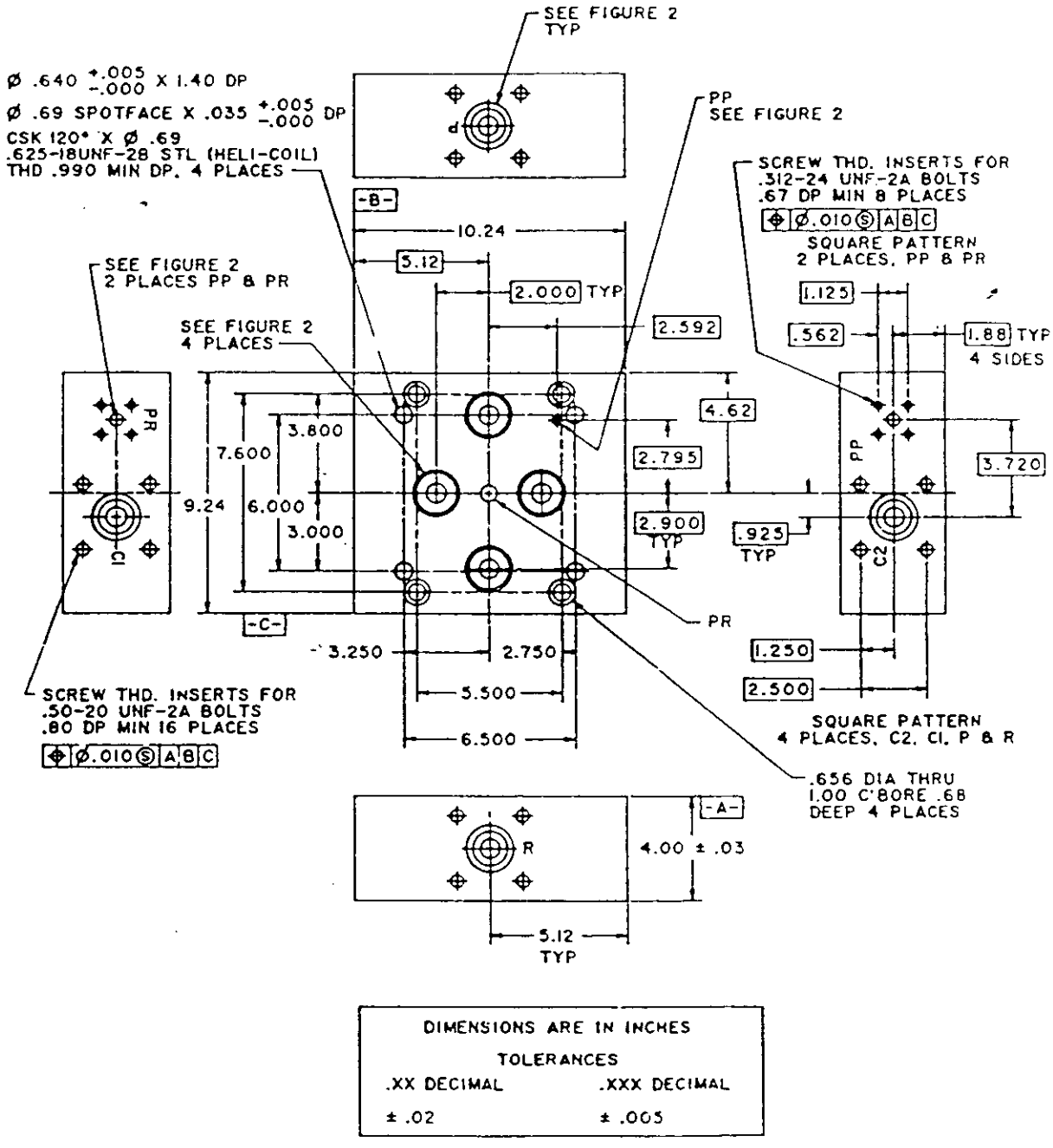
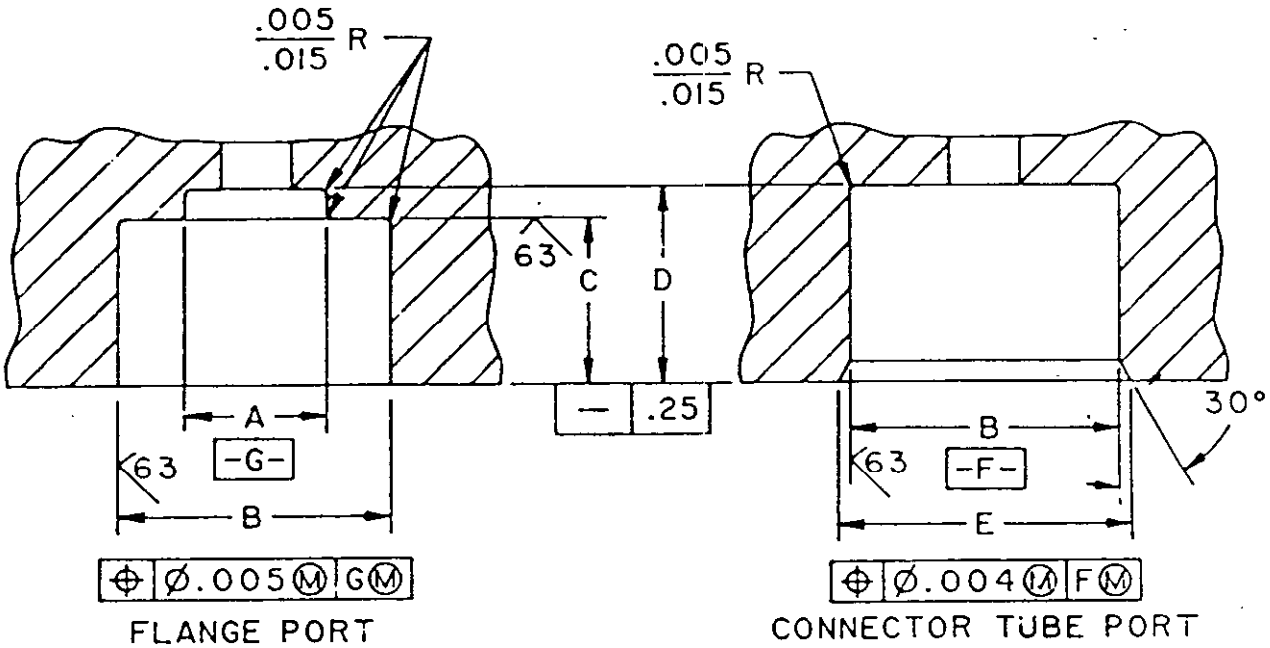


FIGURE 1. *Subplate port locations and interfaces.*

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FEATURE	FOUR-BOLT FLANGE COUNTERBORE DIMENSIONS IN INCHES	
	C1, C2, R & P	PP & PR
A	1.563 +.002 −.000	.438 +.001 −.000
B	2.125 +.002 −.000	.688 +.002 −.000
C	.307 +.005 −.000	.151 +.005 −.000
D	.502 +.010 −.000	.251 +.010 −.000

FEATURE	CONNECTOR TUBE COUNTERBORE DIMENSIONS IN INCHES		
	C1, C2, R & P	PP	PR
B	1.622 +.001 −.000	.361 +.001 −.000	.676 +.002 −.000
D	.250 +.005 −.000	.312 +.006 −.000	
E	1.660 ±.005	.400 ±.005	.720 ±.005

FIGURE 2. *Subplate, valve and flushing block port details.*

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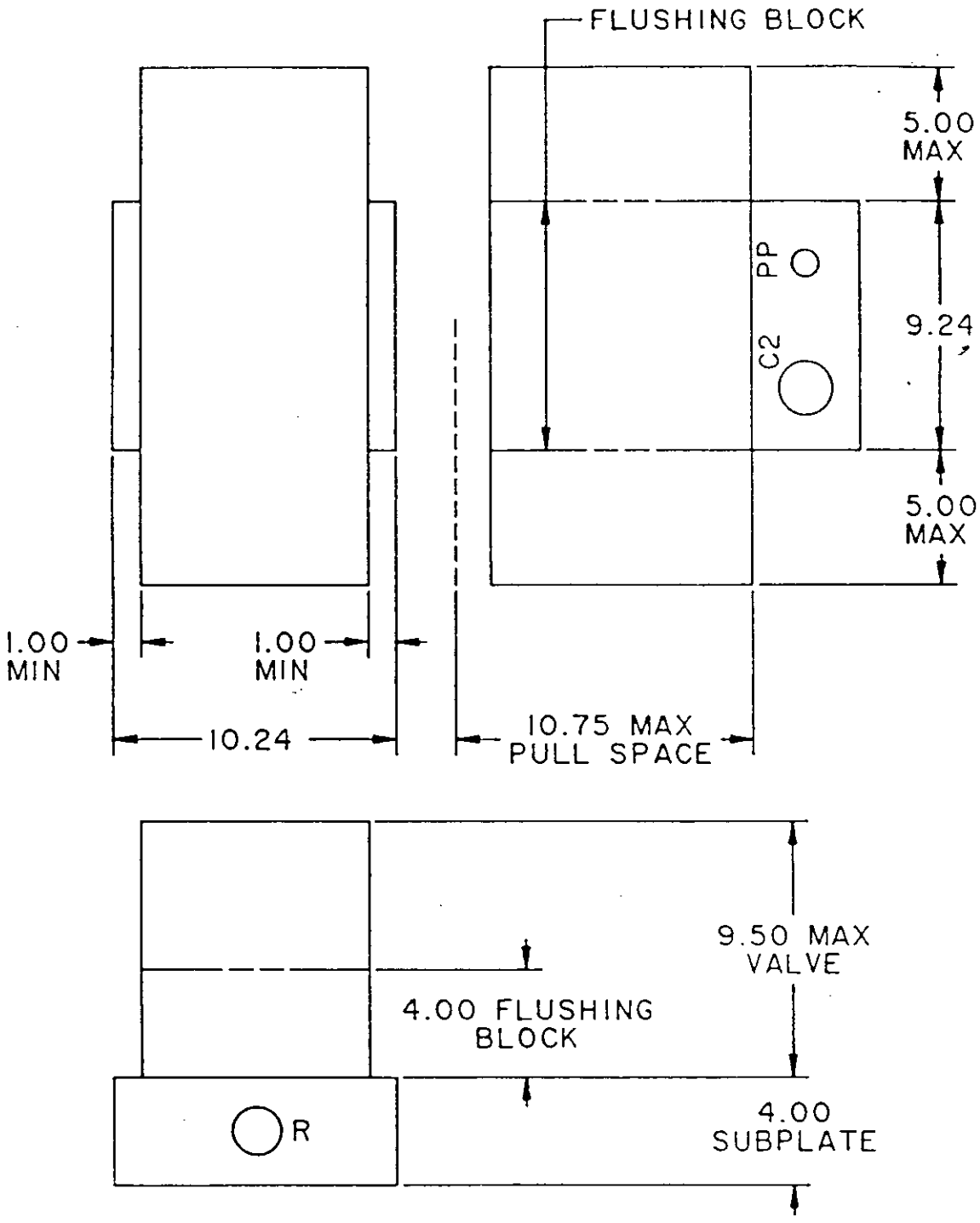
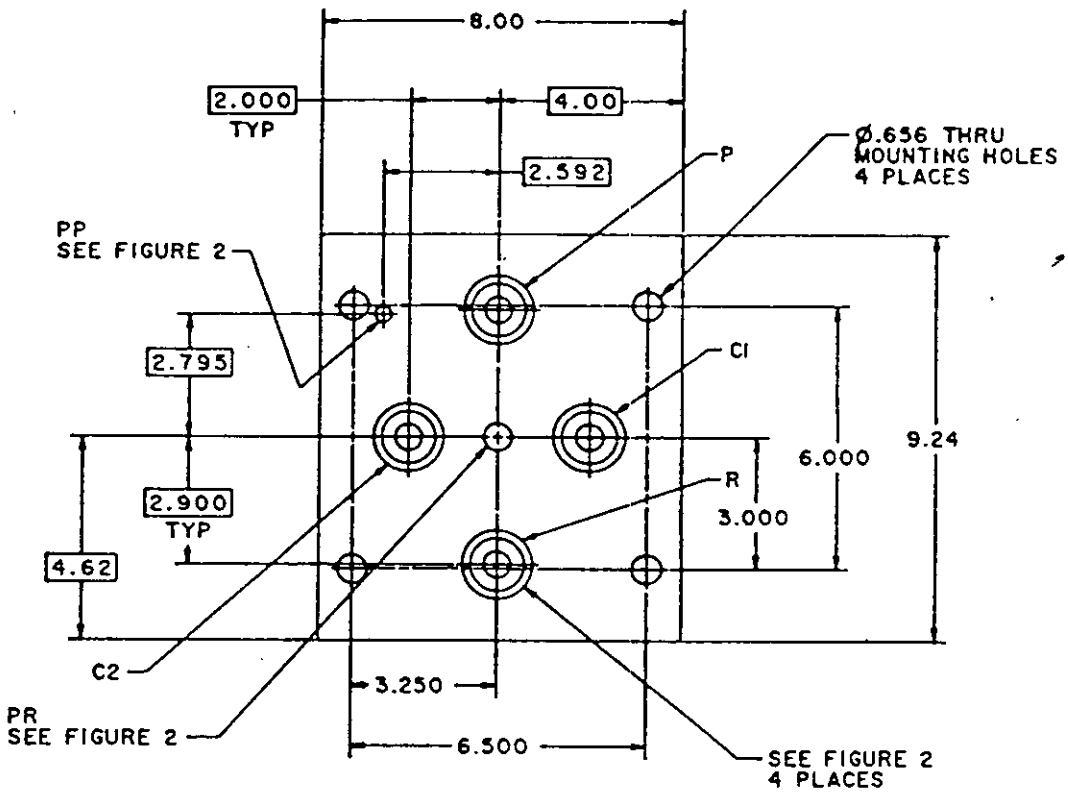


FIGURE 3. *Envelope of valve/subplate assembly or flushing block/subplate assembly.*

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DIMENSIONS ARE IN INCHES	
TOLERANCES	
.XX DECIMAL	.XXX DECIMAL
± .02	± .003

FIGURE 4. *Valve and flushing block port sizes and locations.*

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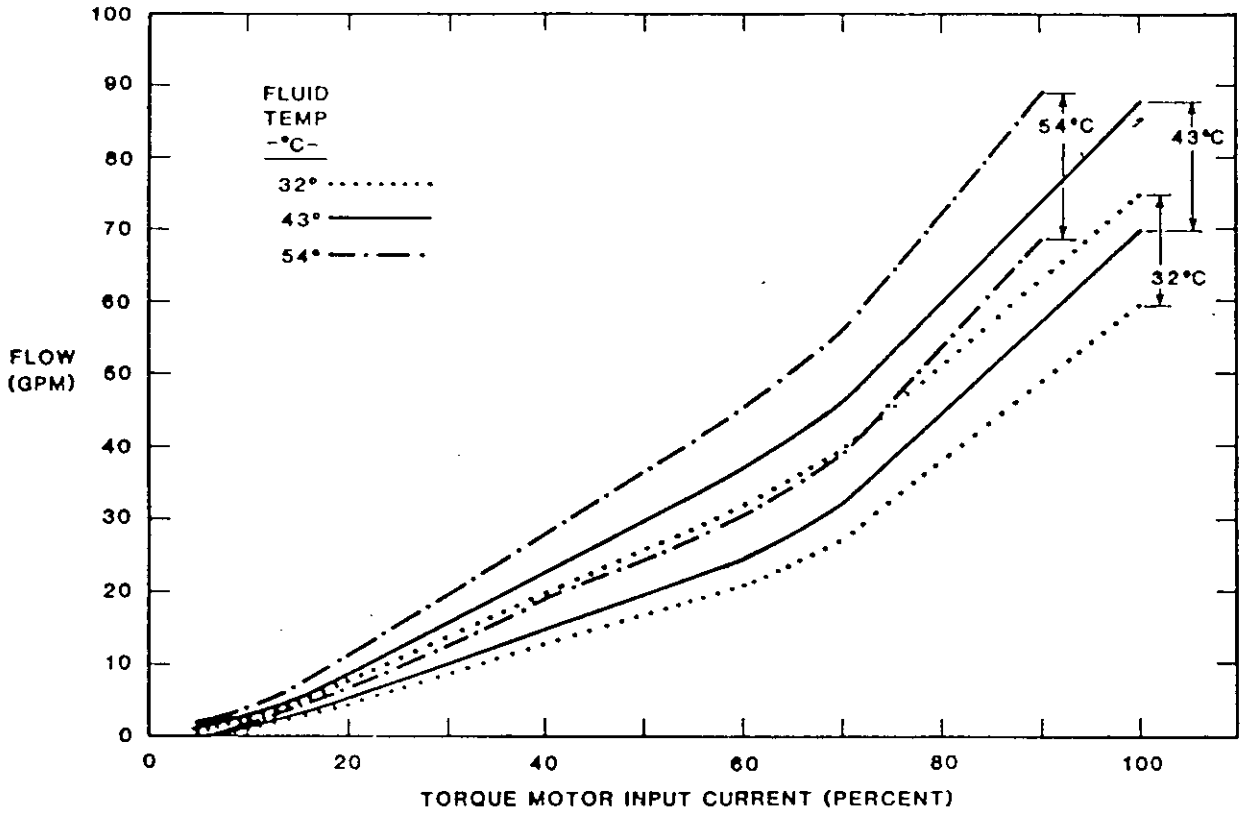


FIGURE 5. *Acceptable – 80 servovalve flow gain limits as a function of fluid temperature.*

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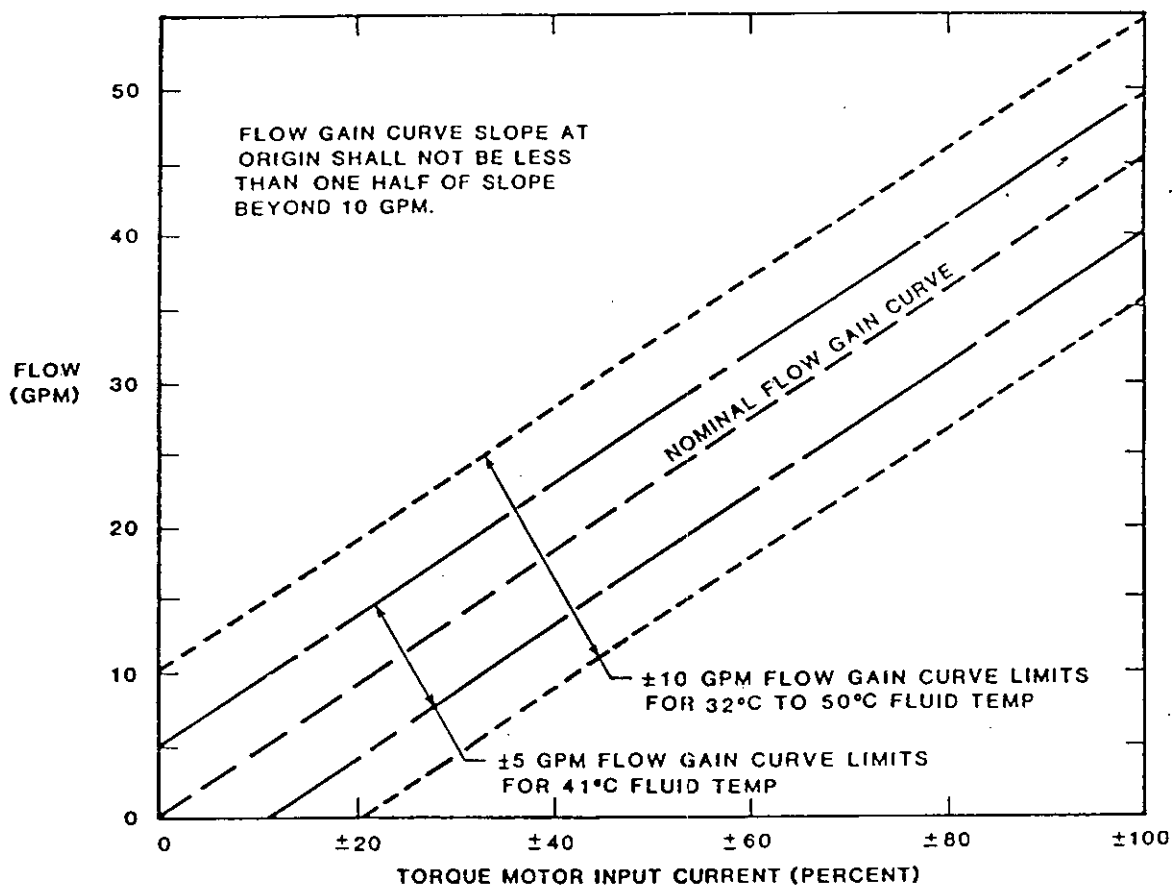


FIGURE 6. *Acceptable −50 servovalve flow gain limits as a function of fluid temperature.*

Preparing activity:
Navy - SH
(Project 4810-N075-01)

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